Camrelizumab versus investigator's choice of chemotherapy as second-line therapy for advanced or metastatic oesophageal squamous cell carcinoma (ESCORT): a multicentre, randomised, open-label, phase 3 study

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Summary

Background **Patients with advanced or metastatic oesophageal squamous cell carcinoma have poor prognosis and few treatment options after first-line therapy. We aimed to assess efficacy and safety of the anti-PD-1 antibody camrelizumab versus investigator's choice of chemotherapy in previously treated patients.**

Methods **ESCORT is a randomised, open-label, phase 3 study of patients aged 18 to 75 years with a histological or cytological diagnosis of advanced or metastatic oesophageal squamous cell carcinoma done at 43 hospitals in China. Eligible patients had an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1, and had progressed on, or were intolerant to, first-line standard therapy. Patients were randomly assigned (1:1) to camrelizumab (200 mg every 2 weeks) or chemotherapy with docetaxel (75 mg/m² every 3 weeks) or irinotecan (180 mg/m² every 2 weeks), all given intravenously. Central randomisation was done using the Randomization and Trial Supply Management system with block size randomly generated as four or six and stratified by disease and ECOG performance status. The primary endpoint was overall survival, assessed in randomised patients who had received at least one dose of treatment. Safety was assessed in all treated patients. The trial is registered with ClinicalTrials.gov, NCT03099382, and is closed to new participants.**

Findings **From May 10, 2017, to July 24, 2018, 457 (75%) of 607 screened patients were randomly assigned to treatment, of whom 228 received camrelizumab treatment and 220 received chemotherapy. As of data cutoff on May 6, 2019, with a median follow-up time of 8·3 months (IQR 4·1–12·8) in the camrelizumab group and 6·2 months (3·6–10·1) in the chemotherapy group, median overall survival was 8·3 months (95% CI 6·8–9·7) in the camrelizumab group and 6·2 months (5·7–6·9) in the chemotherapy group (hazard ratio 0·71 [95% CI 0·57–0·87]; two-sided p=0·0010). The most common treatment-related adverse events of grade 3 or worse were anaemia (camrelizumab *vs* chemotherapy: six [3%] *vs* 11 [5%]), abnormal hepatic function (four [2%] *vs* one [<1%]), and diarrhoea (three [1%] *vs* nine [4%]). Serious treatment-related adverse events occurred in 37 (16%) of 228 patients in the camrelizumab group, and in 32 (15%) of 220 patients in the chemotherapy group. Ten treatment-related deaths occurred, seven (3%) in the camrelizumab group (three deaths from unknown causes, one enterocolitis, one hepatic function abnormal, one pneumonitis, and one myocarditis) and three (1%) in the chemotherapy group (two deaths from unknown causes, and one gastrointestinal haemorrhage).**

Interpretation **Second-line camrelizumab significantly improved overall survival in patients with advanced or metastatic oesophageal squamous cell carcinoma compared with chemotherapy, with a manageable safety profile. It might represent a potential option of standard second-line treatment for patients with oesophageal squamous cell carcinoma in China.**

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See Online for Chinese translation in appendix 1

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Introduction

Oesophageal cancer is the seventh most common malignancy in terms of incidence and the sixth most common leading cause of cancer death worldwide.[1] In 2018, approximately 572 000 new cases of oesophageal cancer were diagnosed and it caused 509 000 cancer deaths globally.[1] Although oesophageal adenocarcinoma is the predominant subtype of oesophageal cancers in non-Asian populations, oesophageal squamous cell carcinoma accounts for 90% of cases in the Asian population.[1,2] Oesophageal squamous cell carcinoma is the fourth most common cancer and the fourth leading cause of cancer-related death in China, and more than half of global oesophageal squamous cell carcinoma

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Research in context

Evidence before this study

We searched PubMed and international conferences (held by the American Society of Clinical Oncology and the European Society for Medical Oncology) for publications of articles and abstracts up to Oct 12, 2019, reporting on immunotherapies in patients with advanced or metastatic oesophageal squamous cell carcinoma using the search terms "esophageal squamous OR oesophageal squamous" and "PD-1 OR PD-L1". This search was limited to articles and abstracts from phase 3 clinical trials. The search resulted in two studies of immune checkpoint inhibitors including pembrolizumab and nivolumab. In the KEYNOTE-181 phase 3 study, an improvement in survival was observed in patients with pembrolizumab versus chemotherapy whose oesophageal squamous cell carcinoma tumours expressed PD-L1 (combined positive score ≥10) as second-line therapy. The ATTRACTION-3 trial reported that nivolumab significantly improved overall survival versus chemotherapy in previously treated patients with advanced oesophageal squamous cell carcinoma. Despite the fact that more than half of global cases of oesophageal squamous cell carcinoma occur in China, no randomised study on immunotherapy for such patients has been reported in China. A phase 1 trial showed manageable safety profile and promising antitumour activity of camrelizumab in Chinese patients with recurrent or metastatic oesophageal squamous cell carcinoma. Therefore, we launched this phase 3 randomised study to further investigate the activity and safety of camrelizumab in this population.

Added value of this study

To our knowledge, ESCORT is the first randomised phase 3 trial to report the superiority of camrelizumab (anti-PD-1 inhibitor) versus chemotherapy for overall survival in Chinese patients with advanced or metastatic oesophageal squamous cell carcinoma who progressed on or were intolerant to first-line chemotherapy. Unlike treatment with nivolumab in ATTRACTION-3, camrelizumab was associated with a reduction in risk of progression or death compared with chemotherapy. Camrelizumab was also well tolerated.

Implications of all the available evidence

The results of this study, along with those from previous studies, supported the use of an anti-PD-1 inhibitor in second-line treatment in oesophageal squamous cell carcinoma. Camrelizumab might represent a new standard second-line treatment option for patients with advanced or metastatic oesophageal squamous cell carcinoma.

cases occur in China.[2,3] Chemotherapy, for example with taxanes, cisplatin, and fluorouracil, is commonly used as the first-line treatment for patients with advanced or metastatic oesophageal squamous cell carcinoma. However, there are few treatment options in second-line setting in patients with advanced or metastatic oesophageal squamous cell carcinoma who progress on or are intolerant to first-line standard chemotherapy.

Several clinical studies have reported promising efficacies and manageable safety profiles of anti-PD-1 antibodies in patients with advanced or metastatic oesophageal squamous cell carcinoma.[4–6] The phase 3 ATTRACTION-3 study[7] showed an improvement in overall survival with nivolumab compared with chemotherapy (median overall survival 10·9 months *vs* 8·4 months; hazard ratio [HR] 0·77, 95% CI 0·62–0·96) in patients with advanced oesophageal squamous cell carcinoma. The phase 3 KEYNOTE-181 study[8] showed an improvement in overall survival in patients with oesophageal squamous cell carcinoma and positive PD-L1 expression (combined positive score [CPS] ≥10) randomised to pembrolizumab compared with those who received chemotherapy (median survival 10·3 months *vs* 6·7 months; HR 0·64, 95% CI 0·46–0·90). Pembrolizumab was approved by the US Food and Drug Administration for patients with recurrent, locally advanced, or metastatic oesophageal squamous cell carcinoma with a PD-L1 CPS of 10 or higher who had disease progression on previous systemic therapy. Given the high prevalence of oesophageal squamous cell carcinoma in China and the shortage of effective treatment options, there is an unmet medical need to develop novel efficacious drugs for patients in China.

Camrelizumab (SHR-1210), a high-affinity, fully humanised, selective IgG4-κ monoclonal antibody against PD-1, has shown activity across a wide range of solid tumours.[9–13] In a phase 1 study, camrelizumab showed promising antitumour activity and a manageable toxicity profile in previously treated patients with recurrent or metastatic oesophageal squamous cell carcinoma.[9] Treatment with camrelizumab resulted in an objective response rate of 33·3% and a median progression-free survival of 3·6 months. On the basis of these data, we did a phase 3 study to assess the efficacy and safety of camrelizumab as second-line therapy in patients with advanced or metastatic oesophageal squamous cell carcinoma.

Methods

Study design and participants

The ESCORT trial was a randomised, open-label, phase 3 study done in 43 hospitals in China (appendix 2 pp 27–28). Eligible patients were aged 18–75 years; had histological or cytological diagnosis of oesophageal squamous cell carcinoma; had advanced, recurrent, or distant metastatic disease; progressed on (or were intolerant to) previous first-line chemotherapy (chemoradiotherapy for recurrent or metastatic settings was included; patients who had tumour progression during or within 6 months after radical chemoradiotherapy or [neo]adjuvant therapy were also eligible; only one patient who was intolerant to

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See Online for appendix 2

first-line chemotherapy was enrolled); had at least one measurable lesion according to Response Evaluation Criteria In Solid Tumors (RECIST, version 1.1); had an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1; had a life expectancy of at least 12 weeks; and had adequate haematological, hepatic, and renal function. Baseline laboratory tests required to assess eligibility included absolute neutrophil count, platelet count, haemoglobin, total bilirubin, alanine aminotransferase, aspartate aminotransferase, and creatinine clearance. Fresh or archival tumour samples were required to be provided for biomarker analyses. The main exclusion criteria included a diagnosis of other malignant tumours (except cured basal cell carcinoma of skin and squamous cell carcinoma of skin, or cervical carcinoma in situ, or breast cancer in situ) within 5 years before the first study dose; CNS metastases; a history of or having active autoimmune disease; a history of anti-PD-1 or anti-PD-L1 therapy; a history of interstitial lung disease or active non-infectious pneumonia with corticosteroids treatment; a congenital or acquired immunodeficiency; active hepatitis B or C viral infection; previous therapies with corticosteroids or other immunosuppressants within 2 weeks before initiation of study treatment; and previous therapies with monoclonal antibodies, chemotherapy, targeted therapy, or radiotherapy within 4 weeks before the start of study treatment. Full eligibility criteria are available in the protocol (appendix 2).

The study was done in accordance with the Declaration of Helsinki and the Good Clinical Practice Guideline. The protocol and all amendments were approved by the institutional review board or independent ethics committee of each study site. All patients provided written informed consent.

Randomisation and masking

Eligible patients were randomly assigned (1:1) to receive either camrelizumab or the investigator's choice of chemotherapy (docetaxel or irinotecan). Randomisation was done using the Randomization and Trial Supply Management system (RTSM system; Bioknow, Beijing, China) via central block randomisation method with the block size randomly generated as four or six and stratified by disease (ie, advanced or metastatic) and ECOG performance status (ie, 0 or 1). The randomisation sequences were generated by a randomisation specialist of the sponsor in SAS (version 9.4). Investigators registered patients at each site via the web-based system and assigned eligible patients on the basis of randomisation sequences obtained from the RTSM system. The sponsor's study team was masked to treatment allocation. Patients and investigators were not masked to treatment allocation.

Procedures

For more on **EORTC QLQ-OES18** see https://www.eortc.org/

Camrelizumab was given intravenously over 30 min at a dose of 200 mg on day 1 of each 2-week cycle. Docetaxel (75 mg/m², on day 1 of each 3-week cycle) or irinotecan (180 mg/m², on day 1 of each 2-week cycle) were given intravenously over 60 min. Treatment was continued until disease progression (defined according to RECIST, version 1.1), unacceptable toxicity, patient withdrawal, or investigator decision, whichever occurred first. In the camrelizumab group, patients who had first disease progression, but were in clinically stable condition based on the investigators' judgment, could continue treatment until progression was confirmed at a subsequent evaluation 4 weeks later. If a patient was not confirmed to have disease progression in a subsequent evaluation, the treatment would be continued; otherwise the treatment would be discontinued unless the investigator felt that the patient would further benefit from the treatment. Treatment interruption was allowed to manage toxic events until the prespecified criteria for treatment resumption were met. Dose reduction of camrelizumab was not permitted. Criteria of dose adjustment and treatment or retreatment for docetaxel and irinotecan was determined by the investigators according to routine clinical practice. Detailed guidelines for dose interruption and modification are available in the protocol.

Tumour response was evaluated every 8 weeks by the investigator according to RECIST version 1.1. The main types of medical examinations routinely done in this study were radiographic tests, such as CT and MRI, as recommended by RECIST version 1.1. Disease progression in this study indicated regional recurrence or distant metastasis. After discontinuation of treatment, patients were followed up every month to assess survival. Adverse events were assessed continuously for up to 90 days after the last dose and graded according to the Common Terminology Criteria for Adverse Events, version 4.03. Adverse events were recorded up to 30 days after the last administration, and treatment-related adverse events were recorded until day 90 after the end of treatment.

Using tumour samples obtained before study treatment, most of which were archival samples except for 58 fresh tissue samples, tumour cell PD-L1 expression was measured by a central laboratory using a human PD-L1 immunohistochemistry kit (6E8 antibody, Shuwen Biotech, Deqing, Zhejiang, China). PD-L1 expression was quantified as tumour proportion score, which was defined as the percentage of viable tumour cells showing partial or complete membrane staining ($\geq 1^{+}$), relative to all viable tumour cells present in the sample.[14] A tumour proportion score of 1% was used as the cutoff of PD-L1 positive and negative as defined in a previous study.[14] Two pathologists were involved in the evaluation. If there was a disagreement between the two pathologists, they discussed it to make a final decision.

Health-related quality of life was assessed using European Organisation for Research and Treatment of Cancer (EORTC) QLQ-C30 and EORTC QLQ-OES18 scales, every 8 weeks from the start of cycle one until 30 days after the last dose. The EORTC QLQ-C30 scale comprises 30 items that are combined to form five functioning scales (physical, role, cognitive, emotional and social), three symptom scales (fatigue, pain, and nausea or vomiting), a global health status quality-of-life scale, and six single-item scales (dyspnoea, insomnia, appetite loss, constipation, diarrhoea, and financial problems).[15] The EORTC QLQ-OES18 scale is mainly used for patients with oesophageal cancer and contains 18 items.

Outcomes

The primary endpoint was overall survival, which was defined as the time from randomisation to death from any cause. Secondary endpoints were progression-free survival (defined as the time from randomisation to the first disease progression or death from any cause), objective response rate (defined as the percentage of patients whose best overall response was complete or partial response), and health-related quality of life (as measured by EORTC QLQ-C30 and EORTC QLQ-OES18). Exploratory endpoints included the associations between efficacy of camrelizumab and PD-L1 expression or other biomarkers and immunogenicity assessments. Some exploratory endpoints will be reported elsewhere once these data are available.

Statistical analysis

Assuming a median overall survival of 7·0 months with chemotherapy,[16–22] 365 events would provide the study with at least 80% power to detect a 2·5-month increase following treatment with camrelizumab (target HR=0·74), with a two-sided log-rank test at a 0·05 level of significance. We expected that approximately 20% of patients would be alive or lost to follow-up, so 438 patients had to be enrolled.

All patients who were randomly assigned and received at least one dose of treatment were included in the full analysis set. Patients who received at least one dose of study treatment constituted the safety set. Efficacy was analysed in the full analysis set and safety was analysed in the safety set. Patient-reported quality of life outcomes were analysed for patients who had a baseline and at least one post-baseline assessment of QLQ-C30 or QLQ-OES18. Median overall survival, progression-free survival, and duration of response were estimated with the Kaplan-Meier method, and 95% CIs were calculated using Brookmeyer-Crowley method.[23] These time-to-event endpoints were compared between treatment groups with the log-rank test stratified by the randomisation strata. For overall survival, a sensitivity analysis was done using the Max-Combo test with Fleming-Harrington (FH) weights of FH (0, 1), FH (1, 0), and FH (1, 1).[24] A post-hoc analysis was done using the rank-preserving structural failure time model to adjust the post-discontinuation anti-PD-1 or anti-PD-L1 treatments in chemotherapy group, and the survival after adjustment was estimated in Kaplan-Meier methods.

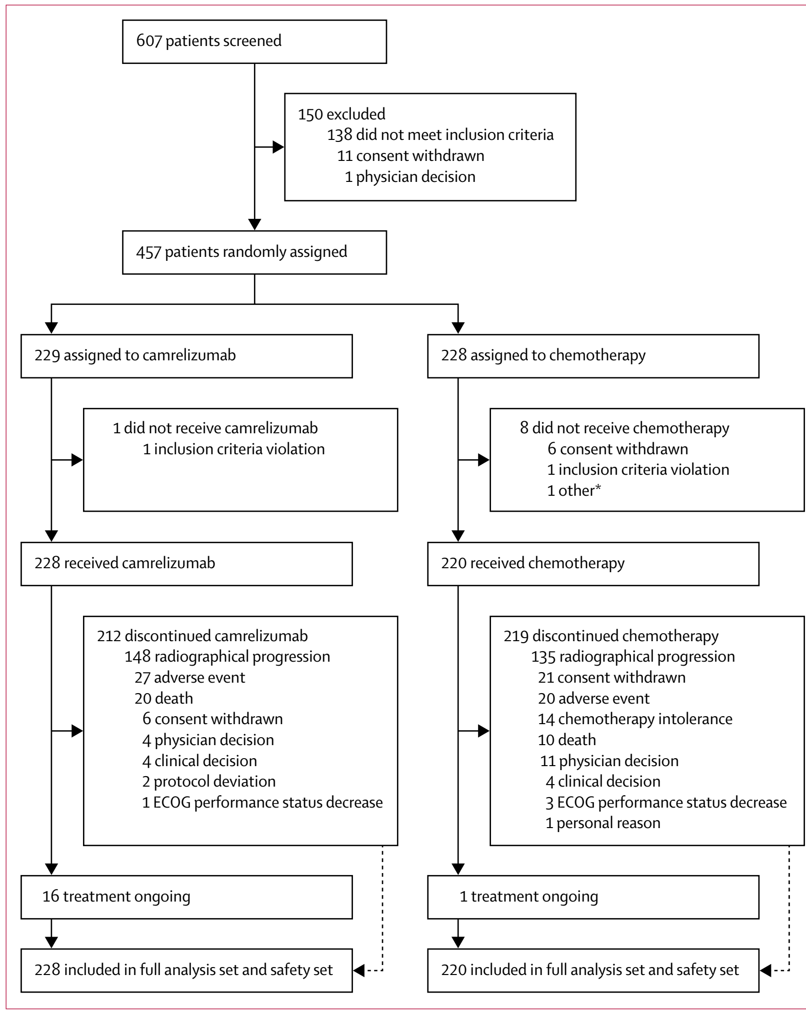


Figure 1: **Study profile**
ECOG=Eastern Cooperative Oncology Group. *Reason cannot be traced.

The HR for overall and progression-free survival and the corresponding 95% CIs were calculated using the stratified Cox proportional hazards model for full analysis set. The subgroup analysis of overall survival and progression-free survival included prespecified analyses of age, sex, ECOG performance status (0 *vs* 1), disease stage, number of involved organs, and PD-L1 expression level. Post-hoc analyses of chemoradiotherapy type, lymph node metastases, and chemotherapy regimens were also included in the subgroup analysis of overall survival. The proportional hazards assumption for overall survival was tested by adding a treatment by time interaction into the Cox model. The HR for overall survival and the corresponding 95% CI for patients who had reactive cutaneous capillary endothelial proliferation compared with patients who did not were also calculated post-hoc using the Cox proportional-hazards model. The analysis of overall survival by PD-L1 status in the forest plot was prespecified. The addition of PD-L1 status to the Cox model (including the interaction

	Camrelizumab group (n=228)	Chemotherapy group (n=220)
Age, years		
Median (IQR)	60 (54–65)	60 (54–65)
Sex		
Female	20 (9%)	28 (13%)
Male	208 (91%)	192 (87%)
ECOG performance status		
0	46 (20%)	44 (20%)
1	182 (80%)	176 (80%)
Histological grade		
Well or moderately differentiated	93 (41%)	84 (38%)
Poorly differentiated	65 (29%)	61 (28%)
Unknown	70 (31%)	75 (34%)
Number of organs with metastases		
1	95 (42%)	84 (38%)
≥2	133 (58%)	136 (62%)
Site of metastases		
Liver	55 (24%)	47 (21%)
Lung	107 (47%)	94 (43%)
Bone	30 (13%)	28 (13%)
Lymph node	189 (83%)	193 (88%)
Other	45 (20%)	37 (17%)
PD-L1 expression		
<1%	129 (57%)	118 (54%)
≥1%	93 (41%)	98 (45%)
<5%	184 (81%)	171 (78%)
≥5%	38 (17%)	45 (20%)
<10%	196 (86%)	181 (82%)
≥10%	26 (11%)	35 (16%)
Previous therapy		
Surgery	113 (50%)	105 (48%)
Radiotherapy	155 (68%)	138 (63%)
First-line platinum-based chemotherapy	218 (96%)	205 (93%)

Data are n (%), unless otherwise specified. ECOG=Eastern Cooperative Oncology Group.

Table 1: **Baseline characteristics**

p value) was a post-hoc analysis. Overall response rate and disease control rate were presented with accompanying 95% CIs calculated on the basis of the Clopper-Pearson exact method. The analyses of overall response rate and disease control rate were done in the full analysis set. These patients who were not assessable were included in the response analysis and counted as non-responders. Differences in patient-reported quality of life outcomes between week 8 and baseline were calculated and compared between two treatment groups via the analysis of covariance model with baseline and randomisation strata as covariates. All analyses were done with SAS (version 9.4). A previously specified interim analysis was cancelled according to an amendment to the protocol on April 12, 2018 (version 5.0). This trial is registered with ClinicalTrials.gov, NCT03099382.

Role of the funding source

The study was designed by the leading principal investigators (JX and JH) and the sponsor. The sponsor contributed to data collection and data interpretation in collaboration with all investigators, funded data analyses, and participated in manuscript preparation with all authors. Medical writing assistance was supported by the funder. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

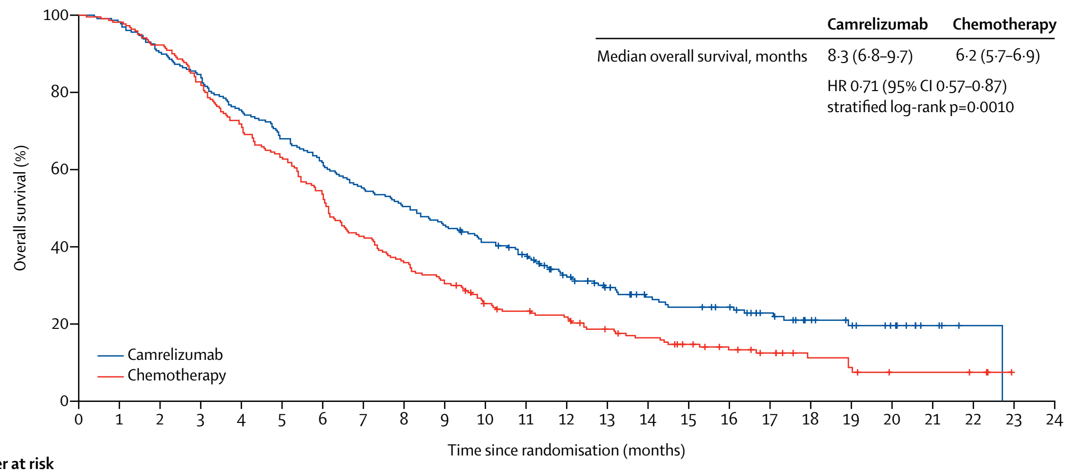
Results

Between May 10, 2017, and July 24, 2018, 457 eligible patients were randomly assigned to treatment, 229 to the camrelizumab group and 228 to the chemotherapy group (figure 1). One patient from the camrelizumab group and eight patients from the chemotherapy group did not receive assigned treatment after randomisation. As a result, 228 patients in the camrelizumab group and 220 patients in the chemotherapy group received at least one dose of assigned treatment and comprised the full analysis set and safety analysis population (table 1, appendix 2 pp 7–8). 43 patients received docetaxel and 177 patients received irinotecan.

As of the data cutoff date on May 6, 2019, the median follow-up duration was 8·3 months (IQR 4·1–12·8) in the camrelizumab group and 6·2 months (3·6–10·1) in the chemotherapy group. 16 (7%) of the 228 patients in the camrelizumab group and one (<1%) of the 220 patients in the chemotherapy group remained on treatment.

172 (75%) deaths occurred in the camrelizumab group and 191 (87%) in the chemotherapy group. The median overall survival was 8·3 months (95% CI 6·8–9·7) in the camrelizumab group compared with 6·2 months (5·7–6·9) in the chemotherapy group (stratified HR 0·71, 95% CI 0·57–0·87; two-sided p=0·0010; figure 2A). Overall survival after adjustment for the post-discontinuation anti-PD-1 or anti-PD-L1 treatment in the chemotherapy arm is shown in the appendix (appendix 2 p 3). 19 patients in the chemotherapy group received anti-PD-1 or anti-PD-L1 therapy after discontinuation of study treatment. For testing of the proportional hazards assumption, the p value of treatment by time interaction was 0·39. To assess the potential delayed treatment effect, the prespecified Max-Combo test also showed a significant increase in overall survival with camrelizumab compared with chemotherapy (two-sided p=0·0008). Kaplan-Meier estimated 6-month overall survival was 63% (95% CI 56·5–69·1) in the camrelizumab group versus 55% (47·7–60·9) in the chemotherapy group. The estimates for 12 months were 34% (95% CI 27·6–39·9) and 22% (17·0–28·1), respectively. Subgroup analysis showed that the survival benefits favoured camrelizumab versus chemotherapy in all tested subgroups (figure 2B; appendix 2 p 4). In patients with a baseline PD-L1 greater or equal to 1%,

A



Number at risk (number censored)

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Camrelizumab	228 (0)	224 (0)	207 (0)	193 (0)	173 (0)	157 (0)	144 (0)	127 (0)	117 (0)	106 (0)	94 (2)	83 (4)	67 (11)	54 (17)	43 (24)	37 (25)	34 (28)	26 (34)	20 (38)	16 (42)	12 (45)	5 (52)	1 (56)	1 (56)	0 (56)
Chemotherapy	220 (0)	216 (0)	203 (0)	182 (0)	160 (0)	141 (0)	120 (0)	94 (0)	81 (0)	71 (0)	56 (3)	47 (5)	44 (6)	35 (8)	29 (10)	24 (12)	19 (16)	15 (18)	10 (23)	9 (23)	5 (24)	4 (25)	4 (25)	1 (29)	0 (29)

B

	Number of deaths/number of patients: Camrelizumab	Chemotherapy	HR (95% CI)
Age at baseline, years			
<65	128/166	139/164	0·75 (0·59–0·96)
≥65	44/62	52/56	0·60 (0·39–0·91)
Sex			
Male	160/208	166/192	0·75 (0·60–0·93)
Female	12/20	25/28	0·45 (0·21–0·93)
ECOG performance status			
0	32/46	31/44	0·93 (0·56–1·53)
1	140/182	160/176	0·67 (0·53–0·84)
Disease stage			
Regional recurrence	32/37	28/33	0·67 (0·40–1·14)
Distant metastasis	140/191	163/187	0·71 (0·57–0·89)
Number of involved organs (metastasis)			
1	71/95	66/84	0·82 (0·58–1·16)
≥2	101/133	125/136	0·63 (0·48–0·82)
PD-L1 expression level			
<1%	100/129	102/118	0·82 (0·62–1·09)
≥1%	68/93	85/98	0·58 (0·42–0·81)
<5%	140/184	146/171	0·74 (0·58–0·94)
≥5%	28/38	41/45	0·51 (0·30–0·84)
<10%	148/196	156/181	0·71 (0·57–0·89)
≥10%	20/26	31/35	0·60 (0·32–1·08)
Lymph node metastases			
Yes	55/73	54/67	0·72 (0·49–1·06)
No	117/155	137/153	0·70 (0·54–0·90)
Chemotherapy regimens			
Camrelizumab vs docetaxel	172/228	36/43	0·73 (0·52–1·07)
Camrelizumab vs irinotecan	172/228	155/177	0·70 (0·56–0·87)
Overall	**172/228**	**191/220**	**0·71 (0·57–0·87)**

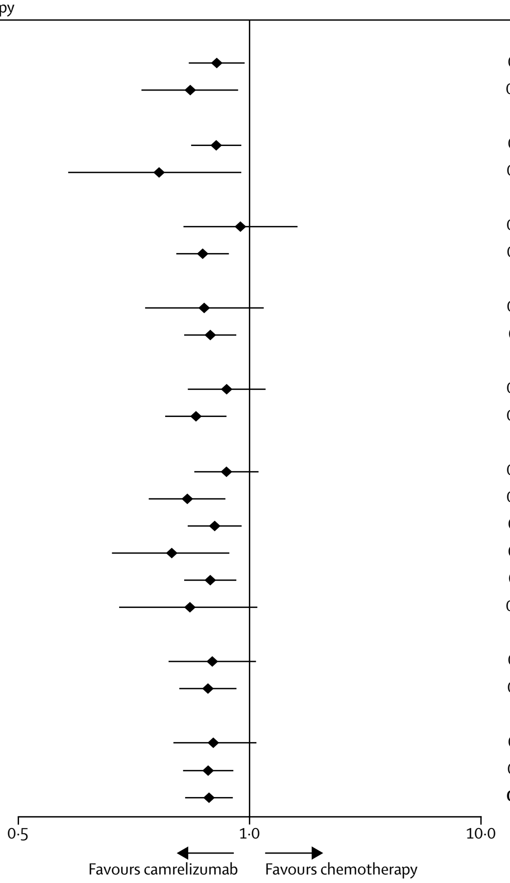


Figure 2: **Overall survival. (A) Kaplan-Meier plot of overall survival. (B) Forest plot for subgroup analyses of overall survival.**
ECOG=Eastern Cooperative Oncology Group. HR=hazard ratio.

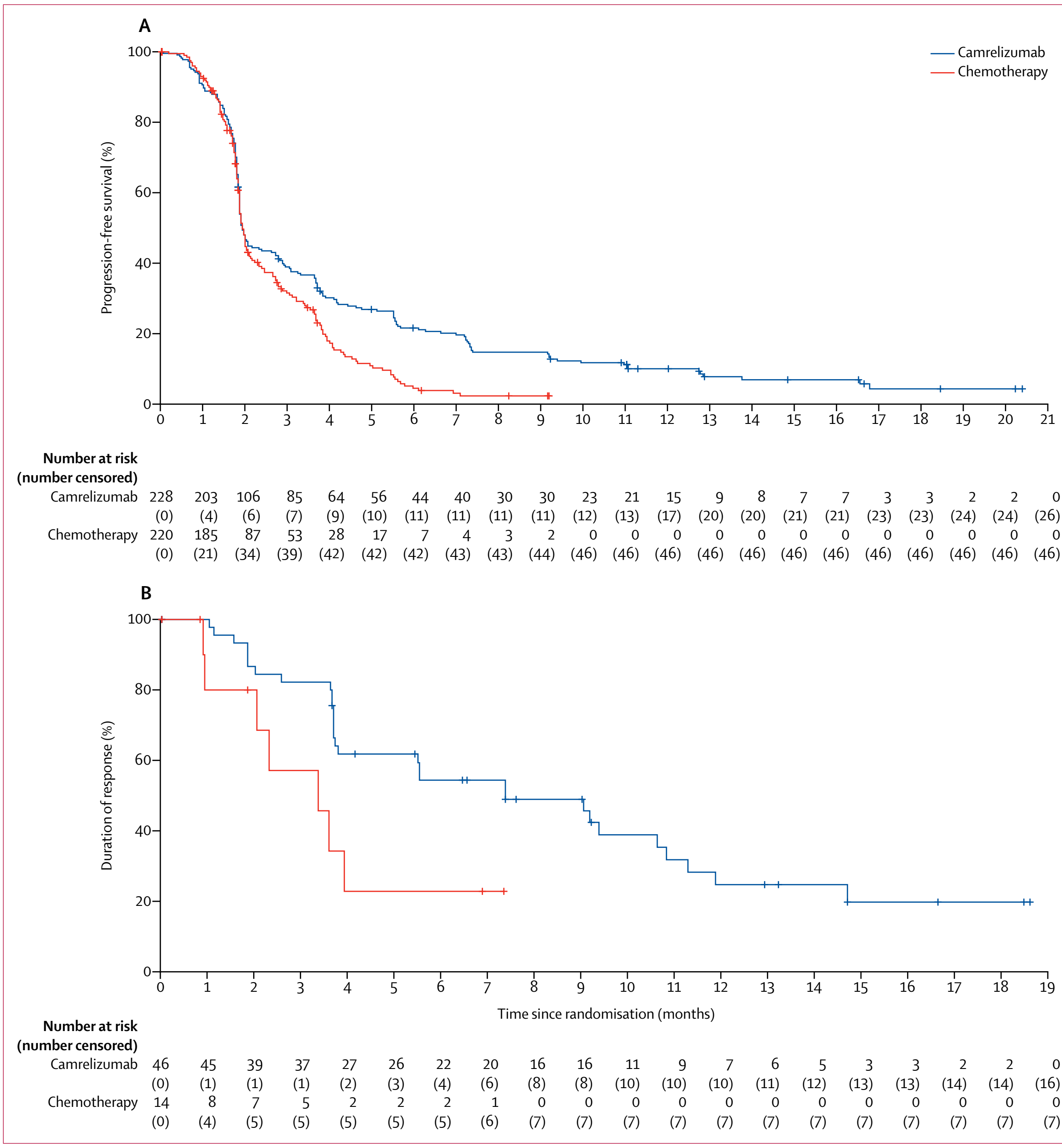


***Figure 3*: Progression-free survival and duration of response. (A) Kaplan-Meier plot of progression-free survival. (B) Kaplan-Meier plot of duration of response.**

the median overall survival was 9·2 months (95% CI 7·0–11·2) in the camrelizumab group and 6·3 months (5·5–7·5) in the chemotherapy group (stratified HR 0·58, 95% CI 0·42–0·81; p=0·0014; appendix 2 p 5). With the addition of treatment with PD-L1 as an interaction term in the stratified Cox model, the p value for the interaction was 0·13, indicating the benefit in overall survival with camrelizumab versus chemotherapy regardless of PD-L1 expression.

202 (89%) of the 228 patients in the camrelizumab group and 174 (79%) of the 220 patients in the chemotherapy group had disease progression or died by the time of data cutoff. Median progression-free survival was 1·9 months (95% CI 1·9–2·4) for patients treated with camrelizumab compared with 1·9 months (1·9–2·1) for those treated with chemotherapy (stratified HR 0·69, 95% CI 0·56–0·86; two-sided p=0·00063; figure 3A). The 6-month progression-free survival estimates were 22% (95% CI 16·4–27·3) in the camrelizumab group and 4% (2·0–8·5) in the chemotherapy group. The estimates for 12 months were 10% (95% CI 6·4–14·6) in the camrelizumab group and not available in the chemotherapy group because patients in this group either developed disease progression, died, or were

censored within 12 months after random allocation. Subgroup analysis showed that the progression-free survival benefit with camrelizumab occurred regardless of subgroup (appendix 2 p 6).

Best overall responses are shown in table 2. Among 46 patients who achieved an objective response in the group receiving camrelizumab and 14 patients in the group receiving chemotherapy, the median duration of response was 7·4 months (95% CI 3·8–10·8) with camrelizumab and 3·4 months (0·9–not reached) with chemotherapy (HR 0·34, 95% CI 0·14–0·92; p=0·017 figure 3B). Response was ongoing in 14 (30%) patients in the camrelizumab group and in four (29%) patients in the chemotherapy group at the time of data cutoff.

The median number of cycles of camrelizumab was six (IQR 4–13), median number of cycles of docetaxel was three (2–3), and median number of cycles of irinotecan was four (2–5).

The primary reason for treatment discontinuation was disease progression in both groups (camrelizumab, 148 [65%] of 228 patients; chemotherapy, 135 [61%] of 220 patients). 249 patients (121 in the camrelizumab group and 128 in the chemotherapy group) received post-discontinuation therapy (appendix 2 pp 9–10), with chemotherapy (82 in the camrelizumab group and 68 in the chemotherapy group) being the most common one. Incidence of treatment discontinuation due to treatment-related adverse events was similar in both groups (16 [7%] of 228 patients in the camrelizumab group *vs* 12 [5%] of 220 patients in the placebo group; appendix 2 pp 11–12). Treatment interruption due to treatment-related adverse events was reported in 30 (13%) of the 228 patients treated with camrelizumab compared with 21 (10%) of the 220 patients who received chemotherapy. Dose reduction in the chemotherapy group occurred in 46 (21%) of 220 patients.

The incidences of treatment-related adverse events were similar in both the camrelizumab and chemotherapy groups (215 [94%] of 228 patients *vs* 198 [90%] of 220 patients; table 3). Treatment-related adverse events of grade 3 or worse occurred in 44 (19%) of 228 patients in the camrelizumab group and 87 (40%) of 220 patients in the chemotherapy group (table 3, appendix 2 pp 13–16). The most common treatment-related adverse events of grade 3 or worse were anaemia (camrelizumab *vs* chemotherapy: six [3%] *vs* 11 [5%]), abnormal hepatic function (four [2%] *vs* one [<1%]), diarrhoea (three [1%] *vs* nine [4%]), asthenia (three [1%] *vs* six [3%]), decreased lymphocyte count (three [1%] *vs* three [1%]), hyponatraemia (three [1%] *vs* three [1%]), and death from unknown causes (three [1%] *vs* two [<1%]).

Serious treatment-related adverse events were reported in 37 (16%) of the 228 patients in the camrelizumab group and 32 (15%) of 220 patients in the chemotherapy group (appendix 2 pp 17–18). Deaths attributed to treatment-related adverse events were reported for seven patients (3%; three deaths from unknown causes, one enterocolitis, one hepatic function abnormal, one pneumonitis, and one myocarditis) in the camrelizumab group and three patients (1%; two deaths from unknown causes, and one gastrointestinal haemorrhage) in the chemotherapy group (appendix 2 pp 19–20).

	Camrelizumab group (n=228)	Chemotherapy group (n=220)
Complete response	1 (<1%)	1 (<1%)
Partial response	45 (20%)	13 (6%)
Stable disease	56 (25%)	62 (28%)
Progressive disease	106 (46%)	103 (47%)
Not evaluable*	1 (<1%)	0
Not assessable†	19 (8%)	41 (19%)
Objective response	46 (20·2%, 15·2–26·0)	14 (6·4%, 3·5–10·5)
Disease control	102 (44·7%, 38·2–51·4)	76 (34·5%, 28·3–41·2)

Data are n (%)or n (%, 95% CI). Responses were evaluated according to Response Evaluation Criteria in Solid Tumors version 1.1 by investigators. *Patient did not have a target lesion at baseline. †Patients who did not have a tumour response evaluation at post-baseline.

Table 2: **Best responses to camrelizumab and chemotherapy treatments**

202 (89%) of the 228 patients in the camrelizumab group had immune-related adverse events, with the most common being reactive capillary endothelial proliferation (182 [80%]), hypothyroidism (44 [19%]), skin reaction (20 [9%]), hepatitis (19 [8%]), pneumonitis (17 [7%]), and hyperthyroidism (14 [6%]).

Despite the high incidence of reactive capillary endothelial proliferation, an adverse event commonly associated with camrelizumab, it occurred in grade 1 (162 [71%]) or grade 2 (19 [8%]), with only one (<1%) patient with a grade 3. The median time to onset of reactive capillary endothelial proliferation was 0·9 months (95% CI 0·8–1·0; derived from Kaplan-Meier analysis). The commonly used therapy to treat reactive capillary endothelial proliferation was local haemostasis (18 [10%] of 182 patients) and surgical resection (11 [6%]). Regression of all reactive capillary endothelial proliferation occurred in 113 (62%) patients. The median time from onset to regression was 6·8 months (95% CI; derived from Kaplan-Meier analysis 5·5–7·7). A post-hoc analysis showed that the median survival of patients who had reactive cutaneous capillary endothelial proliferation and did not have reactive cutaneous capillary endothelial proliferation was 10·1 months (95% CI 8·5–11·5) and 2·5 months (1·9–3·3; HR 0·21, 95% CI 0·15–0·31; appendix 2 p 21).

Health-related quality of life was assessed using the EORTC QLQ-C30 scale in 191 patients in the camrelizumab group and 166 patients in the chemotherapy group, and 190 patients in the camrelizumab group and 165 in the chemotherapy group were assessed with the QLQ-OES18 scale (appendix 2 pp 22–26). Baseline scores on the EORTC QLQ-C30 and EORTC QLQ-OES18 health questionnaires were similar in the camrelizumab and chemotherapy treatment groups (appendix 2 p 22–26). The QLQ-C30 questionnaire showed that patients treated with camrelizumab had a lower risk (calculated by

	Camrelizumab group (n=228)				Chemotherapy group (n=220)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
Any	171 (75%)	34 (15%)	3 (1%)	7 (3%)	111 (51%)	64 (29%)	20 (9%)	3 (1%)
Reactive cutaneous capillary endothelial proliferation	181 (79%)	1 (<1%)	0	0	0	0	0	0
Hypothyroidism	38 (17%)	0	0	0	3 (1%)	0	0	0
Asthenia	19 (8%)	3 (1%)	0	0	59 (27%)	6 (3%)	0	0
Anaemia	18 (8%)	6 (3%)	0	0	69 (31%)	11 (5%)	0	0
White blood cell count decreased	17 (8%)	0	0	0	74 (34%)	29 (13%)	8 (4%)	0
Decreased appetite	11 (5%)	0	0	0	63 (29%)	8 (4%)	0	0
Diarrhoea	10 (4%)	3 (1%)	0	0	59 (27%)	9 (4%)	0	0
Neutrophil count decreased	10 (4%)	0	0	0	42 (19%)	19 (9%)	14 (6%)	0
Lymphocyte count decreased	4 (2%)	3 (1%)	0	0	7 (3%)	3 (1%)	0	0
Nausea	4 (2%)	0	0	0	72 (33%)	9 (4%)	0	0
Vomiting	3 (1%)	0	0	0	52 (24%)	10 (5%)	0	0
Hyponatraemia	1 (<1%)	3 (1%)	0	0	2 (<1%)	3 (1%)	0	0
Lung infection	0	1 (<1%)	0	0	0	4 (2%)	0	0
Febrile neutropenia	0	0	0	0	0	5 (2%)	0	0
Bone marrow toxicity	0	0	0	0	2 (<1%)	4 (2%)	1 (<1%)	0
Death	0	0	0	3 (1%)	0	0	0	2 (<1%)

Data are n (%). Grade 1–2 treatment-related adverse events occurring in ≥10% of patients in either group are listed; grade 3, 4, and 5 treatment-related adverse events occurring in ≥1% in either group are listed.

***Table 3:* Treatment-related adverse events**

observing whether the lower 95% CI was above 0, or the upper 95% CI was below 0, both of which indicated a significant difference) of decreased global health status, decreased functioning (emotional functioning and social functioning), and worsening symptom domains (fatigue, pain, nausea and vomiting, appetite loss, and diarrhea) from baseline to week 8 compared with patients with chemotherapy. All other QLQ-C30 functioning and domain items did not differ between groups. The QLQ-OES18 questionnaire showed that camrelizumab was also associated with a lower risk of deterioration in symptom domains from baseline to week 8 in terms of reflux, taste problems, and coughing problems, compared with chemotherapy. All other QLQ-OES18 functioning and domain items did not differ between groups.

Discussion

This randomised phase 3 study showed that camrelizumab improved outcomes compared with investigator's choice of chemotherapy as second-line therapy in Chinese patients with advanced or metastatic oesophageal cancer after first-line standard therapy. Second-line treatment with camrelizumab resulted in longer overall survival and a higher overall response rate than chemotherapy. Camrelizumab had a manageable safety profile, with a lower incidence of treatment-related adverse events of grades 3 or worse compared with that of chemotherapy. Additionally, the risk of deterioration in quality of life was reduced by camrelizumab compared with chemotherapy.

Two randomised phase 3 studies of anti-PD-1 antibodies reported encouraging clinical efficacy in patients with oesophageal squamous cell carcinoma who progressed on, or were intolerant to, previous chemotherapy. The latest phase 3 study (ATTRACTION-3)[7] of nivolumab versus chemotherapy in 419 patients (who were mainly from Japan and South Korea) with advanced oesophageal squamous cell carcinoma showed that nivolumab had a significantly longer median overall survival (10·9 months *vs* 8·4 months; HR 0·77, 95% CI 0·62–0·96; two-sided p=0·019) than chemotherapy, but had a shorter progression-free survival (1·7 months *vs* 3·4 months) and lower objective response rate (19% *vs* 22%). In a Chinese subgroup of the KEYNOTE-181 phase 3 study,[25] median overall survival was longer with pembrolizumab versus chemotherapy (8·4 months *vs* 5·6 months; HR 0·55, 95% CI 0·37–0·83) in the oesophageal squamous cell carcinoma population. In the current phase 3 study done in China, patients with camrelizumab had a longer overall survival and an objective response three times higher than patients who were assigned to chemotherapy. Moreover, the longer overall survival with camrelizumab than with chemotherapy was observed across multiple subgroups analysed. Although the median progression-free survival between the two groups was similar at data cutoff, the Kaplan-Meier curves separated dramatically thereafter, indicating the treatment effect of camrelizumab compared with chemotherapy. These improvements over chemotherapy were clinically meaningful, and further supported the remarkable activity of PD-1 blockade in patients with advanced oesophageal squamous cell carcinoma.

Subgroup analysis showed that clinical benefits of camrelizumab were observed in all PD-L1 expression subgroups, and patients with higher PD-L1 expression appear to derive more benefit than those with low PD-L1 expression. This finding is comparable to that observed in ATTRACTION-3 study,[7] and suggests that PD-L1 expression might not be a solid biomarker for outcome prediction for oesophageal squamous cell carcinoma. Further validation on the correlation of PD-L1 status and clinical outcomes are warranted.

Median overall survival with chemotherapy in our study was similar to that with other chemotherapy drugs for oesophageal squamous cell carcinoma in the second-line setting. The overall survival reported for docetaxel was 5·5–8·1 months, and for paclitaxel was 6·1–7·2 months in previously treated patients with oesophageal squamous cell carcinoma.[26–28] The ATTRACTION-3 study showed a survival of 8·4 months (95% CI 7·2–9·9) with paclitaxel or docetaxel.[7] The marginal difference of survival in these studies might be caused by economic or medical care access, and patient compliance in different regions.

The safety profile of camrelizumab reported in this study was consistent with that previously seen with camrelizumab in patients with advanced oesophageal squamous cell carcinoma and other tumour types.[9,13,29] No new safety signals with camrelizumab were identified. Potentially treatment-related adverse events were generally manageable with treatment interruption or discontinuation as appropriate, indicating that camrelizumab is well tolerated in patients with oesophageal squamous cell carcinoma. In this study, incidence of treatment-related adverse events of grades 3 or worse in the camrelizumab group was lower than that in the chemotherapy group. The incidence of treatment-related adverse events of grades three or worse reported in patients in the camrelizumab group was similar to that of other PD-1 inhibitors.[6,7] Reactive cutaneous capillary endothelial proliferation, which was observed in 80% of 228 patients who received camrelizumab treatment, has also been reported in other clinical studies of camrelizumab. Reactive cutaneous capillary endothelial proliferation was characterised by a tripartite growth cycle of proliferation, plateau, and involution, which emerged after receiving camrelizumab, and most patients could regress after discontinuation of camrelizumab. A phase 1 study done in patients with advanced hepatocellular carcinoma, gastric, or oesophagogastric junction cancer showed that the incidence of reactive cutaneous capillary endothelial proliferation (9·3%) with treatment of camrelizumab plus apatinib (a VEGF receptor 2 inhibitor) was numerically lower than that with camrelizumab monotherapy in other trials, suggesting that VEGFA–VEGFR2 signalling pathway could be involved in the pathogenesis of reactive cutaneous capillary endothelial proliferation. We speculate that the balance between enhancers and inhibitors of angiogenesis might be disrupted by the camrelizumab-induced activation of immune response, resulting in the proliferation of capillary endothelial cells. In addition, in a prospective observational study that enrolled 98 patients with solid tumours from a phase 1 study of camrelizumab, the patients who developed reactive cutaneous capillary endothelial proliferation had a higher response rate than those without reactive cutaneous capillary endothelial proliferation (28·9% *vs* 0%, p=0·031),[30] suggesting a potential relationship of reactive cutaneous capillary endothelial proliferation with tumour response.

It was reported that, during second-line treatment and beyond, deterioration in quality of life was inevitable because of decreased functioning and worsening symptoms,[31] which was consistent with our observations. In addition, we found the risk of deterioration in quality of life could be reduced when treated with camrelizumab compared with chemotherapy in the second-line setting.

This study has some limitations. Firstly, the subgroup analysis of survival benefits based on the PD-L1 expression still needs confirmation in further studies. Secondly, evaluation of other biomarkers which might be associated with survival is warranted.

In conclusion, camrelizumab displayed superior efficacy to chemotherapy and tolerable safety profile in Chinese patients with advanced or metastatic oesophageal squamous cell carcinoma after first-line standard therapy. The findings from this study suggest that camrelizumab might represent a novel standard second-line treatment option in Chinese patients with advanced oesophageal squamous cell carcinoma.

Contributors

JH and JX contributed equally to this trial. JH, JX, JZ, and QY were responsible for the conception and design of the study. JH, JX, YC, WZ, YZ, ZC, JiaC, HZ, ZN, QF, LL, KG, YL, YB, ZM, XJ, MZ, JianC, ZF, LG, JW, XZ, TL, ZL, LS, and YS contributed to the collection of data. TZ was responsible for directing the statistical analysis. All authors were responsible for the interpretation of data and writing the manuscript, as well as reviewing and approving the manuscript for submission.

Declaration of interests

JH received research funding from Jiangsu Hengrui Medicine. TZ, QY, and JZ are employees of Jiangsu Hengrui Medicine. All other authors declare no competing interests.

Data sharing

Data from individual patients (including data dictionaries) will not be available (appendix 2 p 31). The study protocol is available in the appendix (appendix 2).

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